ESP32-S3 A7672S 4G Module

(4G/GSM/LTE Cat-1, Wi-Fi 802.11 b/g/n, Bluetooth 5 LE, GNSS, BMS, 7.4V Battery Support)

Hardware Version: **2.0**

Overview:

This advanced IoT development board combines the power of the **ESP32-S3-WROOM-1** Wi-Fi + BLE MCU with the **SIMCom A7672S LTE module**, enabling high-performance wireless and cellular IoT applications.

Designed for industrial and outdoor deployments, the board supports:

- LTE 4G connectivity, Supports GSM fallback
- Wi-Fi + BLE
- Dual-cell battery operation
- Solar charging
- MicroSD storage
- Multiple USB-C interfaces
- External antenna support
- GPIOs for multiple sensor types

1. Key Features:

- **ESP32-S3:** Dual-core Xtensa LX7 processor, AI acceleration, 240 MHz, Wi-Fi + BT 5.0.
- **A7672S:** LTE Cat 1 module with GNSS, supports HTTP(S), MQTT(S), FTP, SSL.
- **SIM Card Slot:** Nano SIM for LTE network access.
- **USB-C Port (3x):** One for ESP32-S3 programming, One for ESP32-S3 USB Host, one for LTE Interface.
- **Battery Support:** Battery Holder with onboard charging IC, which can support 7.4V Battery. supports Model 18650 3.7V.
- **GPIO Headers:** Full pin access for sensors, actuators, GPOIO/I2C/SPI/UART.
- **Onboard LEDs:** Power, network status, Battery Charging status.
- **Reset & Boot Buttons**: For flashing and module management
- **Power On/OFF**: On Board Slide switch

2. Applications

- **IoT Devices**: Remote monitoring, asset tracking, smart metering, and connected devices.
- **M2M Communication**: Machine-to-machine communication for industries like automotive, agriculture, and smart cities.
- **Industrial Automation**: Integration with industrial IoT systems requiring cellular connectivity for real-time data transmission.
- **Point-of-Sale Systems**: Devices requiring cellular data and voice communication for payment processing.

3. Technical Specifications

Electrical Characteristics

- **Supply Voltage**: 10 ~ 26V (JST)
- **Typical Operating Current**:
  - Idle mode: 110mA
  - Transmitting (LTE): 308mA (Depend on Band)
  - ESP TX (802.11b, 1 Mbps, @21 dBm): 340mA
  - ESP RX (802.11b/g/n, HT20): 88mA

- **Power Consumption**: Low power operation for extended battery life.

Interface

- **Serial Interface (UART SIMcom)**: For control and data transfer if USB is not used.

Cellular Specifications

- **Frequency Bands**:
  - LTE FDD: B1/B3/B5/B8
  - LTE TDD: B38/B39/B40/B41
  - GSM: 900 MHz/1800 MHz
  - GNSS: GPS, GLONASS, BeiDou, (Supports Active Antennas)
- **Speed**: Up to 10 Mbps
- **GSM/EDGE Support**: Available in fallback mode for regions with limited 4G coverage.

Voice and SMS

- **Voice Support**: voice calling functionality.
- **SMS Support**: Standard SMS and PDU modes.

Environmental Specifications

- **Operating Temperature**: -35°C to +75°C
- **Storage Temperature**: -40°C to +85°C

Wi-Fi Specifications

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Wi-Fi Multimedia (WMM)
- TX/RX A-MPDU, TX/RX A-MSDU
- Immediate Block ACK
- Fragmentation and defragmentation

- Automatic Beacon monitoring (hardware TSF)
- Four virtual Wi-Fi interfaces
- Simultaneous support for Infrastructure BSS in Station, SoftAP, or Station + SoftAP modes Note that when ESP32-S3 scans in Station mode, the SoftAP channel will change along with the Station channel
- Antenna diversity
- 802.11mc FTM

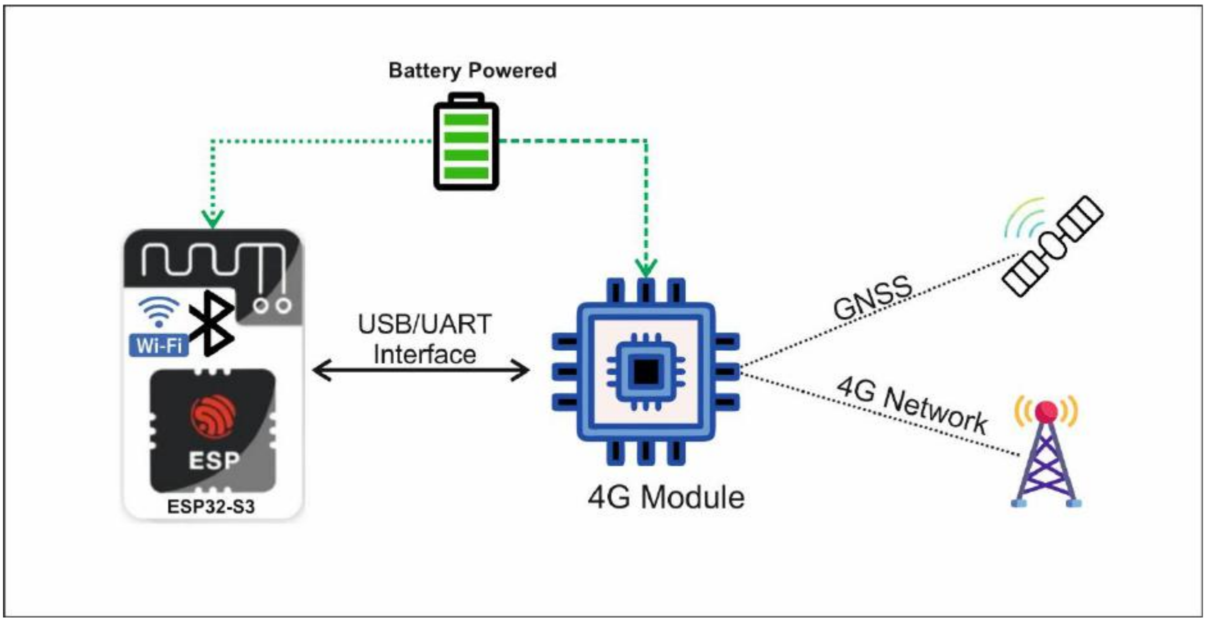
Bluetooth Specifications

- Bluetooth LE: Bluetooth 5, Bluetooth mesh
- High power mode (20 dBm)
- Speed: 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps
- Advertising extensions
- Multiple advertisement sets

CPU and Memory Specifications

- Xtensa® dual-core 32-bit LX7 microprocessor
- Clock speed: up to 240 MHz
- Supported SPI protocols: SPI, Dual SPI, Quad SPI, Octal SPI, QPI and OPI interfaces that allow connection to flash, external RAM, and other SPI devices
- Flash controller with cache is supported
- Power modes: Active, Modem-sleep, Light-sleep, Deep-sleep, Hibernation
- Deep-sleep mode uses 10 μA
- Secure Boot, Flash Encryption, AES, RSA, ECC
- For More Details: https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

Description:



The ESP32-S3 A7672S 4G board integrates ESP32-S3 (Wi-Fi/Bluetooth MCU) with the Simcom 7672S LTE Cat 1 module into a compact development platform ideal for edge IoT applications.

Designed for smart monitoring, remote control, and data acquisition, it supports Wi-Fi, Bluetooth, LTE, and GNSS, offering dual connectivity for flexible deployment in industrial, agricultural, and smart city environments.

Core Modules:

A7672S LTE Module

- LTE Cat 1 modem with fallback to GSM
- Integrated GNSS (GPS, GLONASS, BeiDou)
- Supports TCP/IP, HTTP(S), MQTT(S), FTP
- High-speed data over LTE networks
- Designed for global deployment

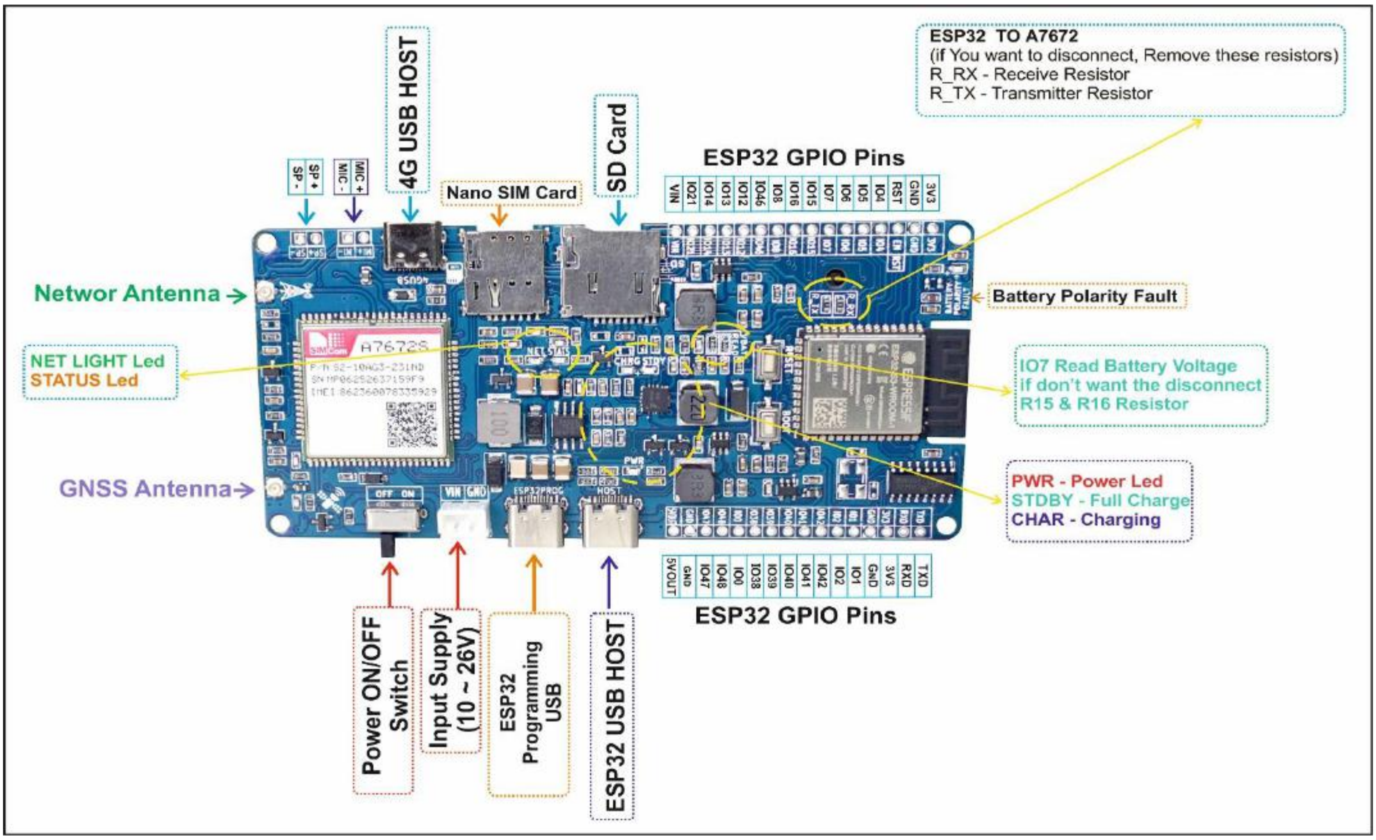
ESP32-S3 MCU:

- Dual-core Xtensa® LX7, up to 240 MHz
- Wi-Fi 802.11 b/g/n, Bluetooth 5 LE
- Rich peripherals: GPIO, ADC, PWM, UART, SPI, I2C
- AI acceleration support
- Ideal for local processing, sensor interfacing, and BLE connectivity

Development Environment

- ESP-IDF (official)
- Arduino IDE
- Visual Studio
- Linux
- Platform IO
- AT command-based control for SIMCOM 7672S
- Supports **OTA firmware updates**

Pinout Details:



Internal connection Details

Device	ESP32-S3 GPIO	Function
A7672S Pin RX	IO17	TX
A7672S Pin TX	IO18	RX
A7672S Pin Power Key	IO10	GSM Power ON
SD Card CS	IO46	SPI
SD Card MOSI	IO4	SPI
SD Card MISO	IO6	SPI
SD Card CLK	IO5	SPI
Battery Voltage	IO7	ADC Read

Pin headers:

ESP32-S3	Connector	Description
3V3	J1	3.3V Output
GND	J1	Ground
EN	J1	Enable (Reset)
IO4	J1	IO4
IO5	J1	IO5
IO6	J1	IO6
IO7	J1	IO7
IO15	J1	IO15
IO16	J1	IO16
IO8	J1	IO8
IO46	J1	IO46
IO12	J1	IO12
IO13	J1	IO13
IO14	J1	IO14
IO21	J1	IO21
VIN	J1	Input Supply
TXD	J2	TXD0
RXD	J2	RXD0
3V3	J2	3.3V Output
GND	J2	Ground
IO1	J2	IO1
IO2	J2	IO2
IO42	J2	IO42
IO41	J2	IO41
IO40	J2	IO40
IO39	J2	IO39
IO38	J2	IO38
IO0	J2	IO0
IO48	J2	IO48
IO47	J2	IO47
GND	J2	Ground
5V OUT	J2	5V Output
IO3 , IO11 , IO45	Pads	IO3 , IO11 , IO45
IO36, IO37, IO35	Pads	IO36 , IO37 , IO35

JST 3	Description
MI+	Mic +
MI-	Mic -

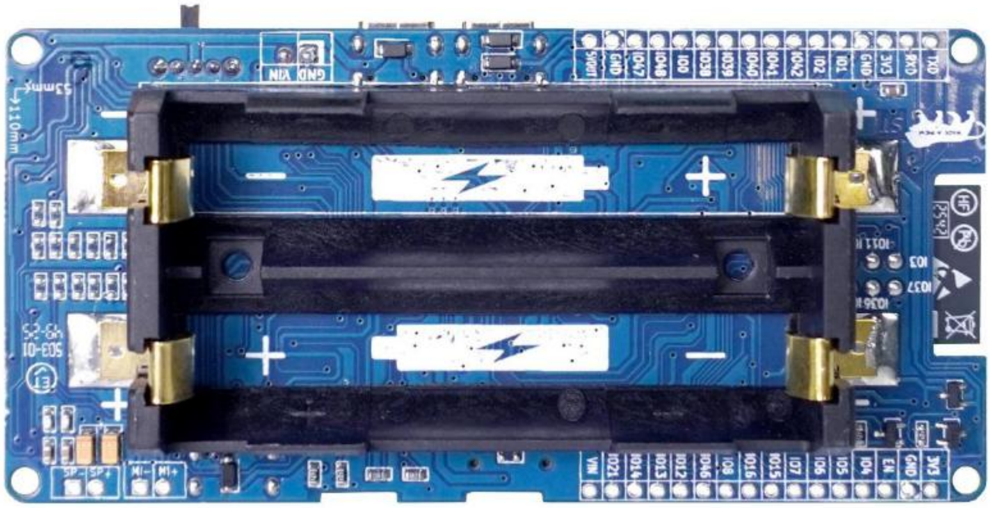
JST 4	Description
SP+	Speaker +
SP-	Speaker -

JST 5	Description
VIN	10 ~ 26V
GND	Ground

GSM to ESP32 Pinout:

A7672S	RX GSM	TX GSM	Power Key
ESP32	IO17 (TX)	IO18 (RX)	IO10

Battery Side : IO7 is used to read ADC data for battery level monitoring.



Battery Selection:

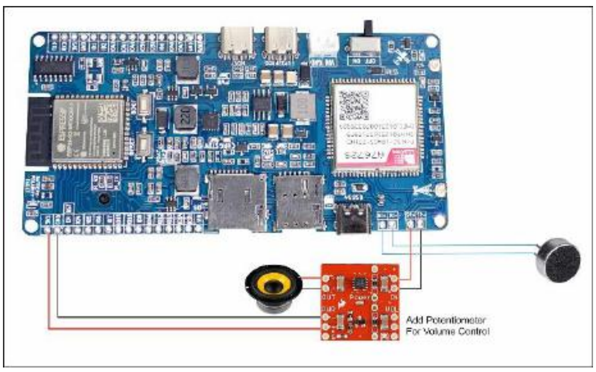
Default 7.4V battery

Model 18650 3.7V Battery

Example Battery

https://sharvielectronics.com/product/lithium-ion-18650-3-7v-2600mah-battery-cell/

Example connection for Speaker:



Solar Panel Charging

The board includes an integrated **solar charging input**, allowing operation in off-grid environments. supporting solar panels with 12~25V voltage input

Applications:

- Outdoor IoT nodes
- Smart agriculture monitoring
- Remote telemetry systems

Storage Expansion

Micro SD Memory Card Socket

An onboard **MicroSD card slot** is provided for:

- Data logging
- Firmware storage
- Offline buffering
- Edge analytics storage

Interface: **SPI**

GPIO Details:

CS	MOSI	MISO	CLK
IO46	IO4	IO6	IO5

Basic AT Commands for A7672 Testing

Module Check

Command	Description	Example Response
AT	Test communication with module	OK
ATI	Get module information (firmware, model)	Module ID & version
AT+CPIN?	Check SIM card status	+CPIN: READY
AT+CSQ	Signal quality	+CSQ: 20,99 (RSSI: 0–31)
AT+CREG?	Network registration status	+CREG: 0,1 (Registered)
AT+COPS?	Current network operator	+COPS: 0,0,"Jio 4G"

Network & IP Setup

Command	Description
AT+CGDCONT=1,"IP","<APN>"	Set PDP context (replace <APN> with your SIM's APN like internet, airtelgprs.com, etc.)
AT+QICSGP=1,1,"<APN>","","",1	Set APN with authentication
AT+QIACT=1	Activate PDP context
AT+QIACT?	Query IP address
AT+QIDEACT=1	Deactivate PDP context

GNSS (GPS) Control

Command	Description
AT+QGPS=1	Start GPS
AT+QGPSLOC=1	Get GPS location
AT+QGPSEND	Stop GPS

HTTP POST Example (Send JSON Data)

Step 1: Initialize HTTP

AT+HTTPINIT

AT+HTTPPARA="CID",1

Step 2: Set URL

AT+HTTPPARA="URL","http://yourserver.com/api/data"

Step 3: Set Content Type

AT+HTTPPARA="CONTENT","application/json"

Step 4: Provide POST Data

Example JSON payload: {"temp":25.6,"humidity":60}

Send length first: AT+HTTPDATA=32,5000

Response:

DOWNLOAD

Now send JSON: {"temp":25.6,"humidity":60}

Expected: OK

Step 5: Execute HTTP POST

AT+HTTPACTION=1

Response:

+HTTPACTION: 1,200,45

OK

(1 = POST)

Step 6: Read Response

AT+HTTPREAD

Step 7: Close HTTP

AT+HTTPTERM

Quick MQTT Test Flow

```
AT+CMQTTSTART
AT+CMQTTACCQ=0,"testclient"
AT+CMQTTCONNECT=0,"tcp://broker.hivemq.com:1883",60,1
AT+CMQTTTOPIC=0,16
sharvi/iot/data
AT+CMQTTPAYLOAD=0,12
{"ok":true}
AT+CMQTTPUB=0,1,60
```

Tips

- Always end commands with \r\n (CRLF).

Refer the Blog for GPS:

https://sharvielectronics.com/gnss-gps-with-quectel-ec200u-4g-lte-gsm-gnss-bt-module/

AT Command Test Code: (Arduino Ide)

```
#define A7672_RX       18
#define A7672_TX       17
#define PW_KEY         10

void setup() {
  pinMode(PW_KEY, OUTPUT);  // Output for Power Key
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, A7672_RX, A7672_TX);  // Connect through UART
  digitalWrite(PW_KEY, LOW); // Enable the A7672 by Power Key

}
void loop() {
  if (Serial.available()) {
    Serial2.write(Serial.read());
  }

  if (Serial2.available()) {
    Serial.write(Serial2.read());
  }
}
```

Send SMS Code: (Arduino Ide)

```
#define A7672_RX 18  // ESP32 pin connected to A7672 TX
#define A7672_TX 17  // ESP32 pin connected to A7672 RX
#define PW_KEY     10

void setup() {
  pinMode(PW_KEY, OUTPUT);
  Serial.begin(115200);             // Monitor
  Serial2.begin(115200, SERIAL_8N1, A7672_RX,  A7672_TX);  // A7672 UART
  digitalWrite(PW_KEY, LOW);
  delay(3000);
  Serial.println("Initializing A7672...");

  sendAT("AT", "OK");
  sendAT("AT+CMGF=1", "OK");  // Set SMS to text mode
  sendAT("AT+CMGS=\"+9199XXXXXXXX\"", ">");  // Replace with your number
  Serial2.print("Hello from A7672!\r");   // SMS content
  Serial2.write(0x1A);  // Ctrl+Z to send
  Serial.println("SMS Sent!");
}

void loop() {
  // Nothing to do in loop
}
// Function to send AT command and wait for expected response
void sendAT(String command, String expectedResponse, unsigned long timeout = 3000) {
  Serial2.println(command);
  unsigned long t = millis();
  String response = "";
  while (millis() - t < timeout) {
    while (Serial2.available()) {
      char c = Serial2.read();
      response += c;
    }
    if (response.indexOf(expectedResponse) != -1) break;
  }
  Serial.println("CMD: " + command);
  Serial.println("RESP: " + response);
}
```

SD Card Example

```
#include <SPI.h>
#include <SD.h>
// MicroSD SPI Pins
#define SD_CS    46
#define SD_MOSI  4
#define SD_SCK   5
#define SD_MISO  6
void setup() {
  // USB Serial Monitor
  Serial.begin(115200);
  delay(1000);
  Serial.println("Initializing MicroSD Card...");
  // Start SPI with custom pins
  SPI.begin(SD_SCK, SD_MISO, SD_MOSI, SD_CS);
  // Initialize SD card
  if (!SD.begin(SD_CS, SPI)) {
    Serial.println("❌ SD Card initialization failed!");
    return;
  }
  Serial.println("✅ SD Card initialized successfully!");
  // Print card type
  uint8_t cardType = SD.cardType();
  if (cardType == CARD_NONE) {
    Serial.println("No SD card attached.");
    return;
  }
  Serial.print("Card Type: ");
  if (cardType == CARD_MMC) Serial.println("MMC");
  else if (cardType == CARD_SD) Serial.println("SDSC");
  else if (cardType == CARD_SDHC) Serial.println("SDHC");
  else Serial.println("UNKNOWN");
  // Print card size
  uint64_t cardSize = SD.cardSize() / (1024 * 1024);
  Serial.printf("Card Size: %llu MB\n", cardSize);
}

void loop() {
}
```

For more GitHub: https://github.com/SharviElectronics/ESP32-S3-A767X-Board/tree/main

Electrical Characteristics:

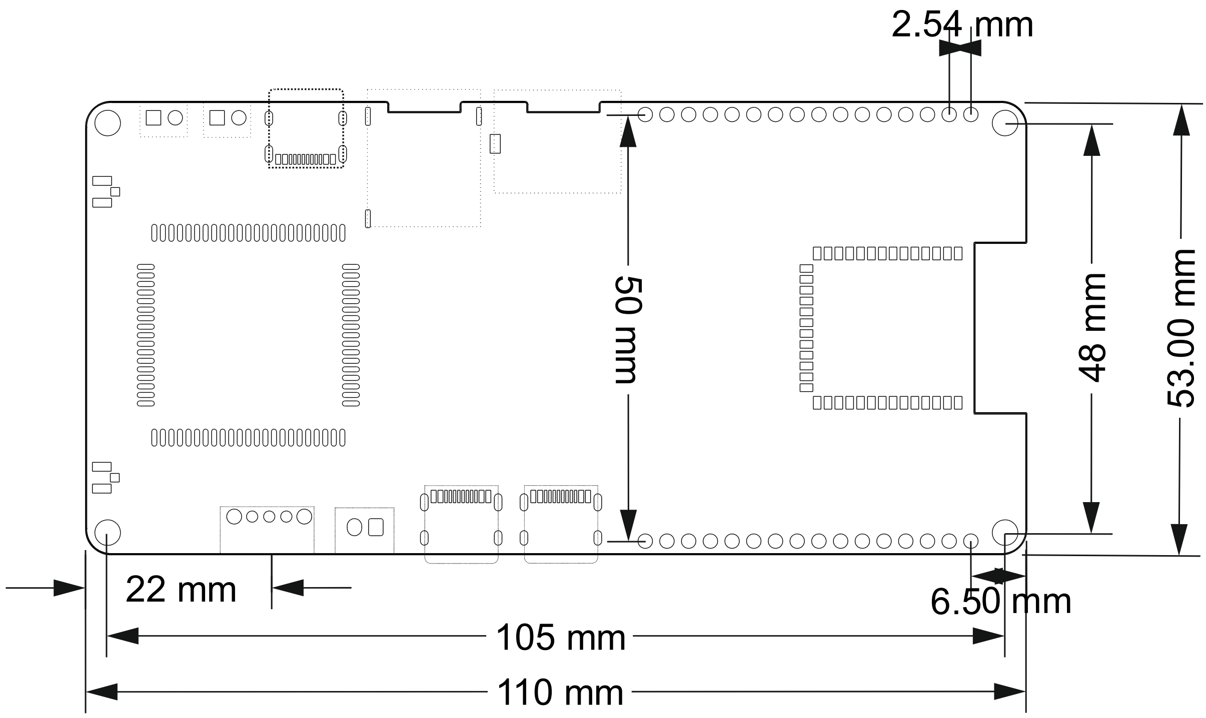
Parameter	Value
Operating Voltage	10V ~ 26V
Operating Current	125mA
Default Baud Rate	115200bps
MCU	ESP32-S3, 240 MHz dual-core (ESP32-S3-WROOM-1-N16R8)
LTE Module	A7672S, Cat 1
Network Protocols	MQTT(S), HTTP(S), FTP, TCP/IP
GNSS Support	Yes (via A7672S)
Wi-Fi	802.11 b/g/n
Bluetooth	BLE 5.0
USB Ports	3x USB-C (2ESP & LTE)
Battery Support	7.4V Li-ion/LiPo with charging
GPIOs	Digital/Analog (via ESP32 header)
Programming	ESP32 via USB-C (UART boot)
Operating Temperature	-35°C to +75°C
Antenna	UFL
SIM Card	Nano Type
Battery Holder	18650
SD Card	MicroSD memory
Battery voltage	ADC read (Io7)

Mechanical Dimensions:

- **PCB Dimensions**: 110 mm x 53 mm
- **Connector Type**: UFL

Dimension Details:

DXF & SVF files are available if Needed contact us.



Notes

- Attach LTE & GNSS antennas before use
- Always insert SIM before powering up
- Use proper Li-ion battery
- Use Proper Powering source

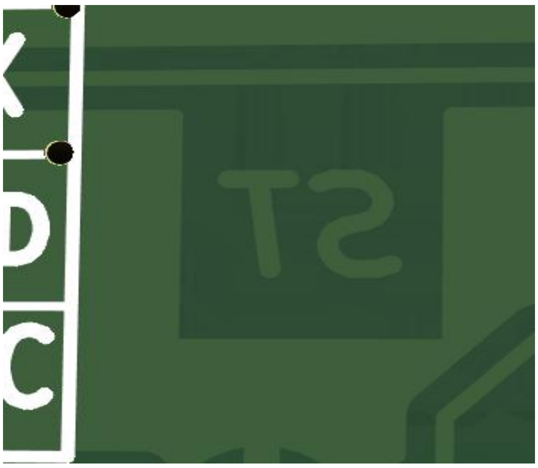
Typical Applications:

- Internet of Things (IoT)
- Industrial Automation
- Point-of-Sale (POS) Systems
- Smart Cities
- Healthcare Applications
- Security and Surveillance
- Agriculture
- Home Automation (Smart Homes)
- Environmental Monitoring
- Consumer Electronics
- Logistics and Supply Chain Management
- Logistics and Supply Chain Management

Ordering Information:

Part Number	Description
ESP32-S3-A7672S-G-C-N	GPS/GNSS, Call, without Solder Header
ESP32-S3-A7672S-G-C-H	GPS/GNSS, Call, with Solder Header
ESP32-S3-A7670C-N-N-N	NO GPS, NO Call without Solder Header
ESP32-S3-A7670C-N-C-N	NO GPS, Call without Solder Header
ESP32-S3-7670G-G-C-N	GPS/GNSS, Call, without Solder Header, **Global Band**

Original Board Marking:



Document Declaration

This datasheet provides technical specifications, details, and performance characteristics of the product(s) described herein. The information contained in this document is provided solely for informational purposes and is subject to change without notice. Every effort has been made to ensure the accuracy of the information in this datasheet. However, no guarantees are made regarding its correctness or completeness.

Product Identification: ESP32-S3-A767XX-X-X-X
Doc Version: Rev1.0
Date: 07-01-2026
Prepared by: R&D Team

Usage and Warranty Disclaimer:

- The information contained within this datasheet is provided on an "as-is" basis. No representations or warranties, either express or implied, are made with respect to the accuracy, completeness, or fitness for a particular purpose of the product or its components.
- The user is responsible for ensuring that the product is used in accordance with its specified conditions and intended applications.
- Sharvi Technologies PVT LTD reserves the right to modify the product and the datasheet content without prior notice.

Copyright:

Limitation of Liability:

In no event shall Sharvi Technologies PVT LTD be liable for any damages arising from the use or inability to use the information provided in this datasheet, even if Sharvi Technologies PVT LTD has been advised of the possibility of such damages.